Semantified CEUR-WS in Wikidata

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Abstract. CEUR Workshop Proceedings (CEUR-WS) has been a non-profit publisher for scientific workshop and conference proceedings since 1995. For more than 30 years it has used HTML and PDF as the single source of truth for the metadata required by indexers such as DBLP and the German national library DNB. Scholars have long craved persistent identifiers for the core entities involved: event series, events, proceedings, papers, editors, authors, institutions, and publishers, with the goal of allowing semantic handling and querying with modern tools.
In this work we show how we enabled the semantification using Wikidata as the target knowledge graph infrastructure. The 2015 Semantic Publishing Challenge queries are finally permanently available with up-to-date metadata. The Scholia portal provides aspect-oriented and user-friendly access and is widely used.
We further point out the challenges of aligning the digital, social, and physical aspects of the ongoing transformation and outline a roadmap to use the available queries to bootstrap a new RDF-based metadata-first publishing approach that will mine metadata from proceedings matter and generate the HTML from it instead of the other way around.

Keywords: Knowledge Graph · Linked Data · Metadata Extraction · Publishing · Semantic Web · Wikidata

1 Introduction

Fig. 1. Most relevant entities for scientific publishing

The Traditional publishing workflows often handle the metadata of the needed core entities (as shown in Figure 1) in an unFAIR way. Findable, Accessible, Interoperable, and Reusable (FAIR) 1.2, Single Point of Truth (SPoT), and Single Source of Truth (SSoT) principles are having low priority. Therefore the opportunities to collect the metadata during the different lifcecyle stages of an event as shown in table 1 are missed. As technical editorr of CEUR-WS we have been feeling this pain for over a decade. This work shows how the HTML/PDF SSoT of CEUR-WS at the ceur-ws.org domain (SPoT) has been used to create a Knowledge Graph representation of the relevant entities in Wikidata between 2022 and 2026 as a step in this direction thus allowing portals such as Scholia to present the RDF query results in an aspect-oriented and user-friendly way.

Table 1. Scientific Event Lifecycle stages

Stage	**Stakeholders**	**Artifacts Created**	**Format**	**Responsibility**
Announcement	Organizers, Marketing Team	Promotional materials, websites, notifications	HTML, Email	Organizers
Call for Papers (CfP)	Organizers, Authors, Reviewers	CfP documents, emails, submission portals, responses	PDF, HTML, Email	Organizers
Performing Conference	Organizers, Chairs, Presenters, Attendees	Schedules, presentations, participant lists, recordings	PDF, PPT, HTML, Video	Organizers
Proceedings Publication	Editors, Chairs, Authors	Finalized papers, indexed and archived content	PDF, HTML	Editors, Authors, Chairs
Indexing	Libraries, Indexing Services	Catalog entries, metadata records	MARC, XML, RDF	Publishers, Libraries
Referencing/Quoting	Academic Community	Citations, references in future works	BibTeX, HTML	Scholars

Digital Traces of Scientific Events during their lifecyle: Table 1 shows the stages of the scientific event lifecycle that cumulatively create artifacts related to the event.

Moving the responsibility for curating data for relevant entities (see Figure 1) to an earlier stage and thus to the event organizers, while semi-automating the process has several benefits. The quality of the data will increase, and its availability will be much earlier, thus influencing FAIRness—e.g., ensuring computer-readable versions of the core entities are available for downstream systems at each step in the lifecycle. Involving a community such as Wikidata in open source style will further enhance the result [4].

Scientific communication increasingly relies on the use of knowledge graphs, as exemplified by Google Scholar[5] and the evolution of Microsoft Academic Graph [20] into OpenAlex [19]. Legacy systems based on technologies such as MARC [14], PICA [6] or XML are also adapting, highlighted by DBLP's adoption

[4] And will create follow-up issues for the single point of truth data to be discussed later

[5] https://scholar.google.com/

of RDF/SPARQL, with the QLever DBLP SPARQL endpoint[6] [3] being the most up-to-date.

1.1 CEUR Workshop Proceedings

Since 1995 CEUR Workshop Proceedings (CEUR-WS) (https://ceur-ws.org/) has published more than 4,200 Volumes and over 90.000 papers as a free online service for publishing proceedings of scientific workshops (and smaller conferences) in computer science, operated by at team of unpaid volunteers.

Technically, all data and metadata of the proceedings are directly represented as a static website using HTML, PDF and HTTP and FTP protocols as provided by the workshop editors along the CEUR-WS "how to submit" guide [4].

The metadata is available after some delay of weeks or month through indexing services such as dblp [16] and K10plus[21][7] and at the archive of the TIB `https://www.tib.eu/`.

After multiple unfinished attempts to make the metadata of the CEUR-WS platform available for semantic analysis and querying we report on the successful use of wikidata and scholia as the knowledge graph platform to provide FAIR metadata.

Doing so was harder than one would expect - the devil is in the details.

1.2 Making CEUR-WS FAIR

The FAIR principles [15,22] with Persistent Identifiers (PIDs) and rich metadata as core components are being adopted by an increasing number of publishing organizations.

To make CEUR-WS metadata fair a set of relevant queries should be supported, as derived from the original set of queries of the 2014 Semantic publishing challenge as outlined in Section **??**. The metadata should be provided following an ontology that is based on established ontologies. The manual and automatic curation of entries should be possible with public access for all stakeholders, e.g., editors, authors, organizers, publishers and indexers. The infrastructure should be stable and there should be sufficient trust in its long term availability An open source non-profit infrastructure is preferred since this is also the mode of operation of CEUR-WS

Given the HTML/PDF/text input of CEUR-WS, the corresponding Knowledge Graph needs to be created and the single-point-of-truth computer readable metadata be separated from the different representations such as HTML so that the above requirements are fulfilled.

Both the HTML and PDF encoding of the original scientific content are structured for the purpose of optimizing the display / output on paper or screens; therefore, there is a structure loss compared to what was originally available in the text document processor files that the authors might have been using [7].

[6] https://qlever.cs.uni-freiburg.de/dblp

[7] `https://dblp.org/`, `https://opac.k10plus.de`

Most scholars are not aware of this loss in the daily use of published content just because the documents are optimized for display and human consumption [24]. For the metadata extraction and use in knowledge graphs the difference is sometimes disastrous, e.g., when a simple text from a PDF document can not be extracted any more due to exotic styling and formatting or simply because only a scanned graphic image version of an older document is available that needs optical character recognition to extract the textual content.

The metadata needed for creating the knowledge graph is only available in natural language/text form and follows rules that have been changed multiple times during the history of CEUR-WS. From 2013 to 2026, there have been 33 different versions of the index file template with no proper tracing of what was changed from version to version. The index and volume HTML files were often edited manually, leading to minor, undocumented differences even within these versions. The usage of the different page versions follows a long-tail Zipfian distribution with 5 versions covering 60% of all volumes.

1.3 Contributions of this Paper

The core contributions presented in this work are:

1. To provide the tools, infrastructure and approaches to modify the CEUR-WS publication workflow to consistently supply high quality FAIR metadata (Semantification) as detailed in section **??**.
2. Splitting the metadata for the three main entities Event, Event Series and Proceedings as a major step towards improving the metadata quality and FAIRness.
3. Providing a bootstrapping [2,5] approach to shift the single point of truth from the current HTML/PDF files to a RDF ready ontology based format.

These contributions are valuable for other publishing use cases see section **??**.

2 From Semantic Publishing Challenge 2015 via Text2KG2023 to Production 2026

3 CEUR-WS Semantification in Wikidata

3.1 Overview

Figure 2 gives an overview of the CEUR-WS Semantification workflow.

4 In-Use: Community Impact - Scholia, WikiCite, CEUR-WS

Scholia [17] is a portal serving wikidata content for the needs of scholars. It is based on a set of over 380 named parameterized SPARQL queries organized

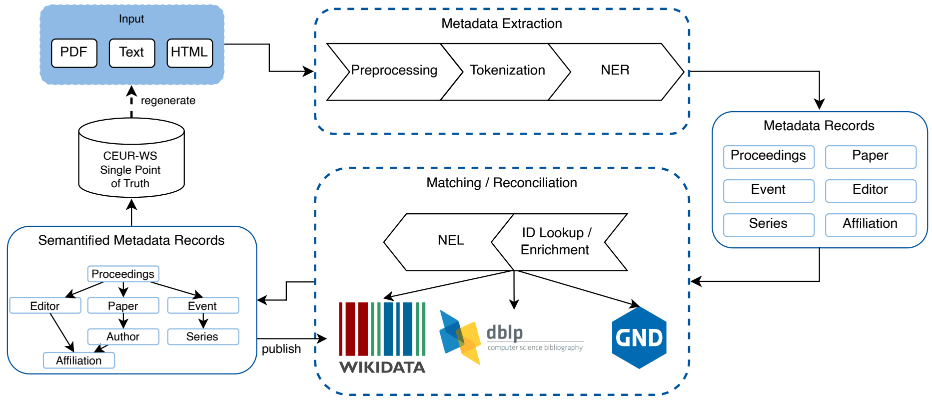

Fig. 2. Workflow of the CEUR-WS Semantification

around 40 different aspects. The Wikidata Scholarly Articles Subgraph Analysis [1] showed that roughly half of the 17 billion triples that form the main graph belong to scholarly articles and their linked entities. The Wikimedia foundation needed to perform a graph split due to a 4 Terrabyte maximum storage limit of the Blazegraph backend in use. The Scholia community decided to use QLever [3] as the new backend and make all queries SPARQL1.1 compliant [23]. Note that before the graph split there was a hold on adding more scholarly articles to Wikidata which was the reason we did no mass-insert CEUR-WS paper metadata yet.

5 Single Point of Truth Bootstrapping

6 Lessons Learned

6.1 Queries as first-class citizens

6.2 Property creation as a community process

Wikidata properties such as P11633 (colocated with)[8] undergo a Wikidata's property proposal process[9] process before a new propery is created. As described in [12] the creation of property proposal[10] took some effort and lead to the clarification that the property is asymmetric. The property was considered as highly relevant and well defined.

[8] https://www.wikidata.org/wiki/Property:colocated with
[9] https://www.wikidata.org/wiki/Wikidata:Property_proposal
[10] https://www.wikidata.org/wiki/Wikidata:Property_proposal/colocated_with

6.3 Legal aspects of Publishing Personal Data of Scholars

The proposed workflow calls for publishing personal data of authors and editors early in the life cycle of an event. According to the legal advisors we asked this is legal since a combination of conditions (Art. 6 GDPR) is met: Explicit consent is given by the stakeholders (Art. 6(1)(a)), the legitimate interest of the stakeholders is pursued (Art. 6(1)(f)), the data might be public already, the processing is necessary for scientific research (Art. 6(1)(e), Art. 89). The strongest condition is that there might be a legal obligation (Art. 6(1)(c)) as would be the case if publishing via major print publishers that have to provide a copy of the proceedings to the German National Library according to German law. Consent, Public Data, and Contractual Necessity are crucial for scholarly publishing.

6.4 Iterative introduction beats big bang

The challenge in introducing the new CEUR-WS workflow is to find a sweet spot between effort and benefit [18]. Given the long-tail distribution of issues of ever more corner and exotic cases that have shown up in the analysis of the existing data and workflow there is potentially a huge list of "small" problems that would require an inadequately high amount of effort to be solved systematically. The "sweet spot" is in solving only the most relevant problems and not fixing some of the exotic details at all or doing it manually. Therefore we will avoid a "big bang" introduction, which would replace the old approach in one single big (and therefore very risky) step. Instead we propose to do smaller changes gradually. As a first step, the main index.html shall be split by years (with a link to the old legacy complete list). Then, the per-year list may be generated from the single point of truth data. This will be continued iteratively, going from the current year back to a point where it seems not reasonable any more to spent further effort.

7 Roadmap: Metadata first Publishing

8 Conclusion

8.1 Supplemental Material Statement

Source code for the prototypes and semantification tools is available via the repositories of the github ceurws organization[11]. The LaTeX sources, bibliography, build pipeline and the `AGENTS.md` governance policy for AI-assisted editing of this paper are available at [13]. The cited software artefacts are released under Apache-2.0: `ceur-spt` [10], `pyCEURmake` [9], `snapquery` [11], and `wdgrid` [8].

[11] https://github.com/ceurws

8.2 Acknowledgements

We would like to thank Jakob Voß for helping with the k10plus matching and creating the Wikidata property colocated with (P11633)[12] in due time.

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This paper is dedicated to the memory of CEUR-WS board member Ralf Klamma † January 2023.

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8.3 Declaration of use of Generative AI

The authors used AI-based coding assistants during the engineering and preparation of this work. Use of such assistants is governed by an AGENTS.md policy file maintained in this paper's repository and, equivalently, in the source repositories of the software artefacts ceur-spt[10] and pyCEURmake[9]. The policy mandates:

- a plan-and-confirm protocol: the assistant must state its analysis, plan, and scope, and receive explicit human confirmation before any read, write, or execution;
- a strict scope for the manuscript itself: AI assistance is limited to spelling/grammar corrections and formatting support (tables, figures, build automation). AI was not used to generate, rewrite or contribute original scientific content, arguments, or prose;

All scientific claims, interpretations and conclusions are the sole responsibility of the authors.

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